Minecraft's The Mace

Assumption: For this whole thing, we assume the player is Steve- an idealized rigid object, not a soft human made of bones, loose joints, and tissues that act like springs.

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1. HULK, SMASH

The mace deals more damage the higher you fall from. Realistically, this *would* be possible. It wouldn't look like an actual hammer or mace in real life, because you need a weapon designed to take all your kinetic energy, channel it completely through the weapon, and dump it straight into your opponent to demolish them.

Hypothetically, you could build a one-handed or two-handed weapon that does this. But Minecraft got the actual physics completely backward.

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2. Let Me Mace Him From Below

The shape isn't even the real issue, that's just a game design choice. The real problem is: **why can't we deal massive damage when we're flying horizontally, or launched high in the air?**

Kinetic energy doesn't care about direction:

$$KE = \frac{1}{2}mv^2$$

If we get launched at high speeds, we should be able to hit someone above us and deal the exact same damage as if we were falling at that speed. But in Minecraft, your weapon does literally shit.

Same thing with the Elytra. If I'm flying at ridiculous speeds horizontally and slam a weapon into someone, they should be decapitated. But because the game only checks for the `FALLING` condition, you have to swap your Elytra for a chestplate before you hit them. Great design, Minecraft.

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3. Fall Damage? What's that?

In-game, you take zero fall damage if you hit someone with the mace. This is why it isn't just a normal hammer. If you jumped off a cliff with a regular hammer in real life and hit someone, sure, you'd transfer energy and destroy them, but **you would still die of fall damage or crash into the ground.**

So for the mace to work, it has to transfer **ALL** your kinetic or gravitational potential energy into your opponent.

Instead of hitting the ground and all that energy destroying *you*, it has to fully release into the opponent and destroy *them* while keeping you alive.

Scientifically, that is basically impossible. We don't have any technology that can transfer kinetic energy into something else with 100% efficiency. Energy is always lost to heat, sound, and material deformation. The future might find a way, but we can't fathom that tech yet.

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4. Here, Hold My Skeleton

Even if the mace magically transferred 100% of the energy, **you would still get fucked and die to death.**

You are going from terminal velocity down to a full stop in less than a second.

$$a = \frac{\Delta v}{\Delta t}$$

If you stop dead in mid-air in a fraction of a second, the deceleration (*G*-forces) is insane. It doesn't matter if you hit the dirt or hit an enemy 'cause your internals would still get completely shredded.

To survive, you can't just stop instantly. You would need to decelerate over an insanely long time while you're inches above the ground:

$$s = \frac{v^2}{2a}$$

To stop safely at low *G*-forces, you need physical distance (s) that you simply do not have when you're right next to the dirt.

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5. The Ideal Gas I Mean Mace (And Why It's Stupid)

To stop your body from getting shredded mid-air, you would basically have to fall backwards to cancel out your forward inertia.

That means the mace would have to:

1. Generate its own localized gravitational field, or
2. Directly manipulate spacetime around you using negative mass/exotic matter to cancel your inertia.

We don't know if negative mass even exists, and controlling gravity or spacetime is tech that is decades or centuries away.

With all of that, sure, hypothetically we *could* make a real-life mace. But that really BEGS to ask,

1. assuming we have all that tech, why the FUCK are we making a melee weapon that relies on our own kinetic energy to do damage, and
2. WHY would we put it on a fucking stick??? Like I don't care what the stick's made of, it's STUPID.

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6. The End (Poem)

So I GUESS the mace can technically be justified as some bullshit hyper-advanced, ridiculously over-engineered sci-fi that makes zero practical sense, but on the other hand, WHAT'S WITH THE WIND CHARGE?! This is just some pure fantasy bull, it's LITERALLY just magic. You cannot trap wind in a portable ball that launches you into the air and also saves you from fall damage and some other bullshit that's gonna make this whole doc even longer and I'm just like whatever you already know what the wind charge does and it's ABSOLUTELY PHYSICALLY IMPOSSIBLE.